



# **Leukemia: Introduction**

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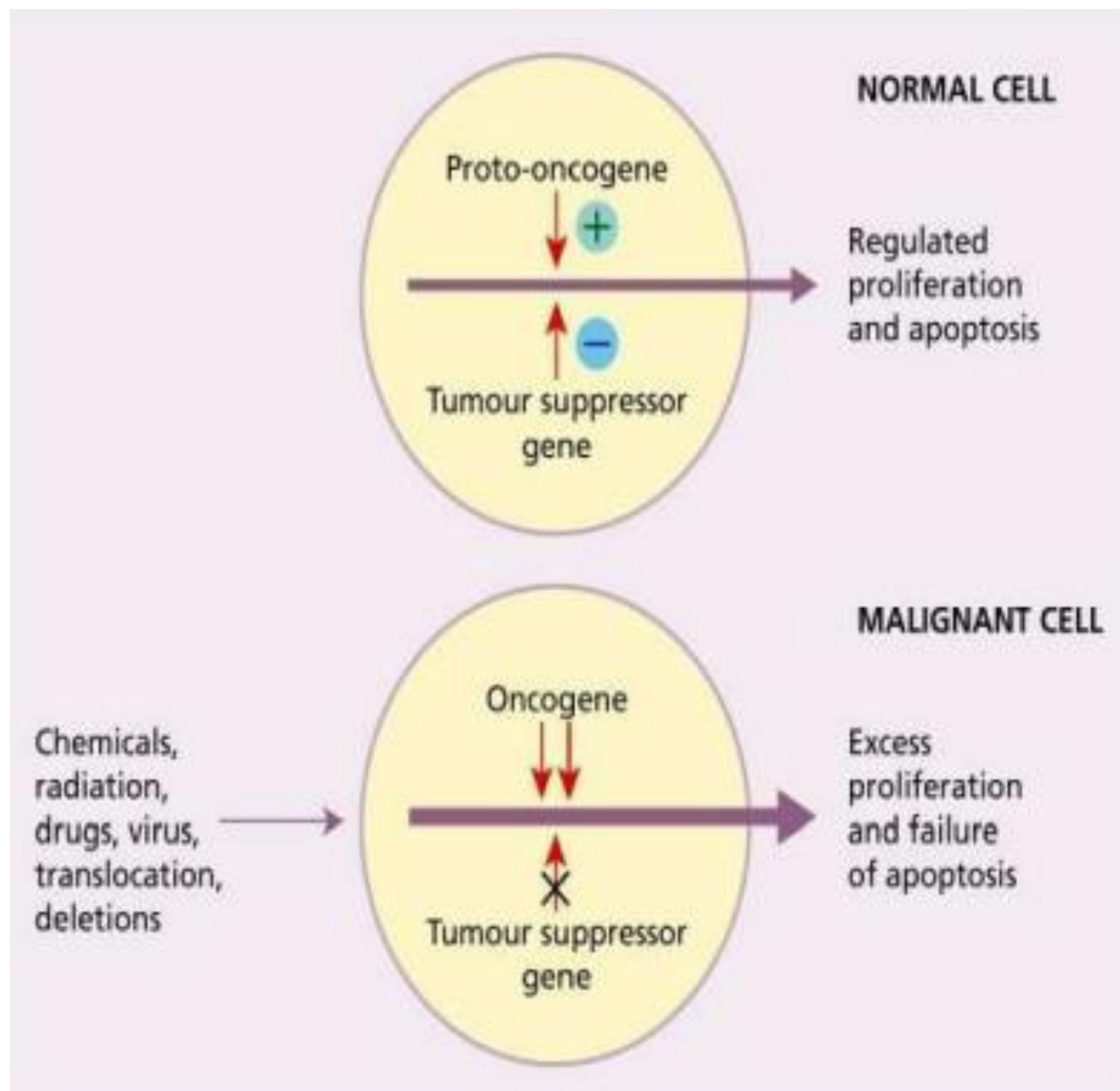
- Neoplasm: an abnormal growth of tissue that, when forming a mass, is commonly referred to as a tumor or tumor.
- Neoplasms are either benign or malignant.
- The malignant neoplasms are known as cancers.
- **Cancer** is a group of diseases involving abnormal cell growth with the potential to invade or spread to other parts of the body.
- Not all tumors are cancerous; benign tumors do not spread to other body parts.
- Cancer is fundamentally a disease of tissue growth regulation failure.

- For a normal cell to transform into a cancer cell, the genes that regulate cell growth and differentiation must be altered.
- The affected genes are divided into two broad categories:
  1. Oncogenes are genes that promote cell growth and reproduction.
  2. Tumor suppressor genes are genes that inhibit cell division and survival.
- Malignant transformation can occur through the formation of novel oncogenes, the inappropriate over-expression of normal oncogenes, or the under-expression or disabling of tumor suppressor genes.
- Typically, changes in **many** genes are required to transform a normal cell into a cancer cell.

# Hematological Malignancies

- Neoplastic diseases of the hematopoietic system (hematopoietic malignancies) are clonal diseases thought to derive from a single cell (clone) in the marrow or peripheral lymphoid tissue that has undergone a genetic alteration.
- They include: Leukemias, myelodysplastic syndromes (MDS), lymphomas, myeloproliferative disorders (MPD), and plasma cell disorders.
- For most leukocyte neoplasms, causes directly related to the development of the malignancy are unknown.

- Most malignancies of the hematopoietic system are acquired genetic diseases, which means that most patients are not born with the illness, but acquire it sometimes later.
- Mutations accumulation involves two groups of genes:
  1. Proto-oncogenes (gain-of-function mutations → oncogene).
  2. Tumor-suppressor genes (loss-of-function mutations),  
(e.g. mutation in p53 of Tp53 gene → leukemia and lymphoma).
- Other causes of hematopoietic malignancies include: toxins, radiation exposure, and benzene.
- Cancer treatment may include chemotherapy, radiation, and/or surgery.



# Leukemias

- Leukemias are a clonal proliferation of malignant leukocytes that arise initially in the BM before disseminating to the peripheral blood, lymph nodes, and other organs.
- Leukemias are a group of malignant diseases of hematopoietic tissue, characterized by the replacement of normal bone marrow elements with abnormal (neoplastic or malignant) cells and abnormal proliferation of blood cells, usually white blood cells, accumulating in the bone marrow and blood and resulting in high numbers of abnormal white blood cells.

- These white blood cells are not fully developed and are called leukemia cells.
- The exact cause of leukemia is unknown.
- Both inherited and environmental factors are believed to be involved.
- Risk factors include smoking, ionizing radiation, some chemicals (such as benzene), prior chemotherapy, and Down syndrome.

**They are broadly classified by:**

1. The type of blood cell giving rise to the clonal proliferation (cell lineage: lymphoid or myeloid).
2. The maturity of the cell or the clinical course of the disease. (Acute: immature malignant cells, Chronic: mature cells are predominant)

➤ **The four main leukemia categories are:**

- I. Acute myeloblastic or myeloid leukemia (AML) (Adults)
- II. Acute lymphoblastic or lymphoid leukemia (ALL) (Children)
- III. Chronic myeloid or myelogenous leukemia (CML) (46-53 years)
- IV. Chronic lymphocytic leukemia (CLL) (elderly > 65 years)

➤ These abnormal cells cause symptoms because of:

1. Bone marrow failure (i.e. anemia, neutropenia, thrombocytopenia).
2. Infiltration of organs (e.g. liver, spleen, lymph nodes, meninges, brain).

➤ Treatment may involve chemotherapy, radiation therapy, targeted therapy, and bone marrow transplant, in addition to supportive care and palliative care as needed.

## Four major kinds of leukemia

| Cell type                                                   | Acute                                                  | Chronic                               |
|-------------------------------------------------------------|--------------------------------------------------------|---------------------------------------|
| Lymphocytic leukemia<br>(or "lymphoblastic")                | Acute lymphoblastic<br>leukemia (ALL)                  | Chronic lymphocytic<br>leukemia (CLL) |
| Myelogenous leukemia<br>("myeloid" or<br>"non-lymphocytic") | Acute myelogenous<br>leukemia (AML or<br>myeloblastic) | Chronic myelogenous<br>leukemia (CML) |

# Acute Leukemia

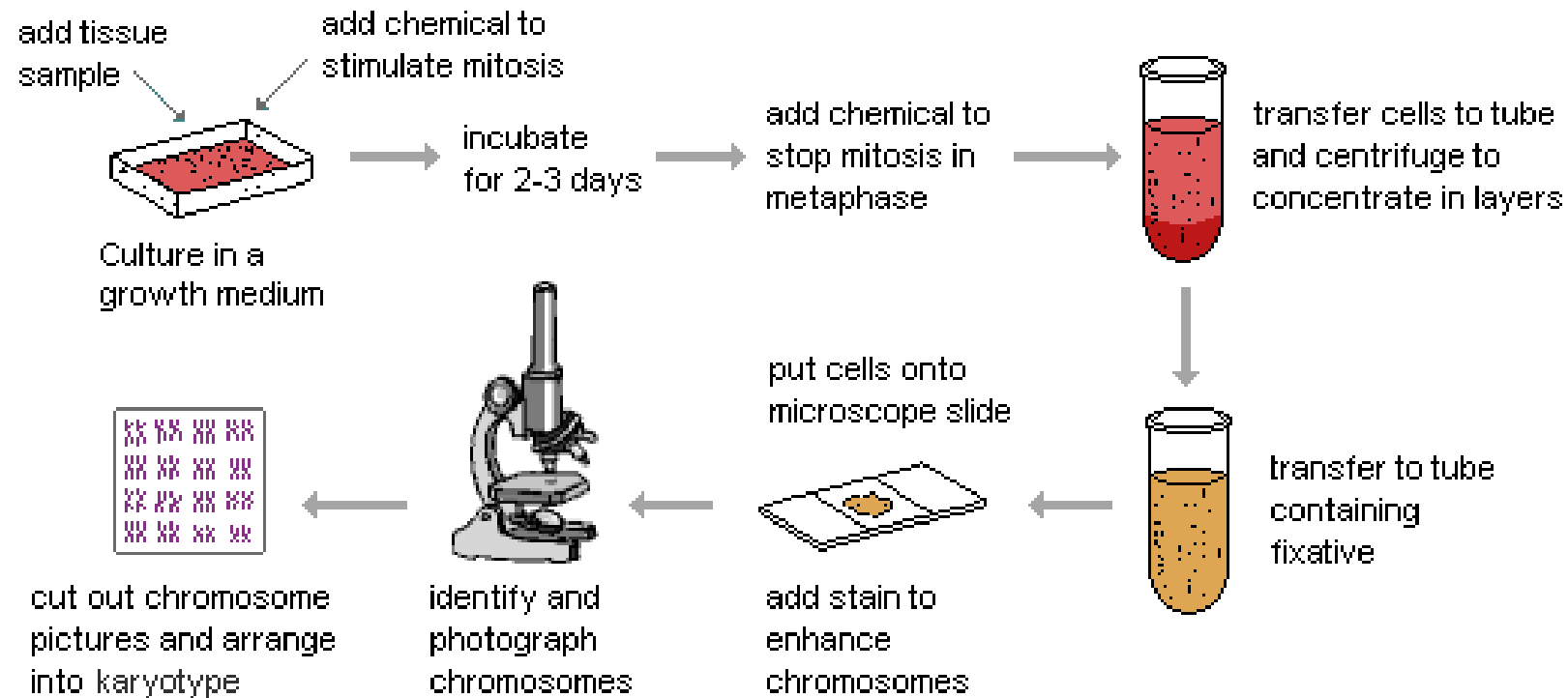
- It is the type of leukemia that is characterized by the malignant proliferation of the immature leukocytes that are called blasts.
- Acute leukemia refers to the rapid, clonal proliferation in the bone marrow of lymphoid or myeloid progenitor cells known as lymphoblasts and myeloblasts, respectively.
- When the proliferation of blasts overwhelms the bone marrow, blasts are seen in the peripheral blood and the patient's symptoms reflect suppression of normal hematopoiesis.

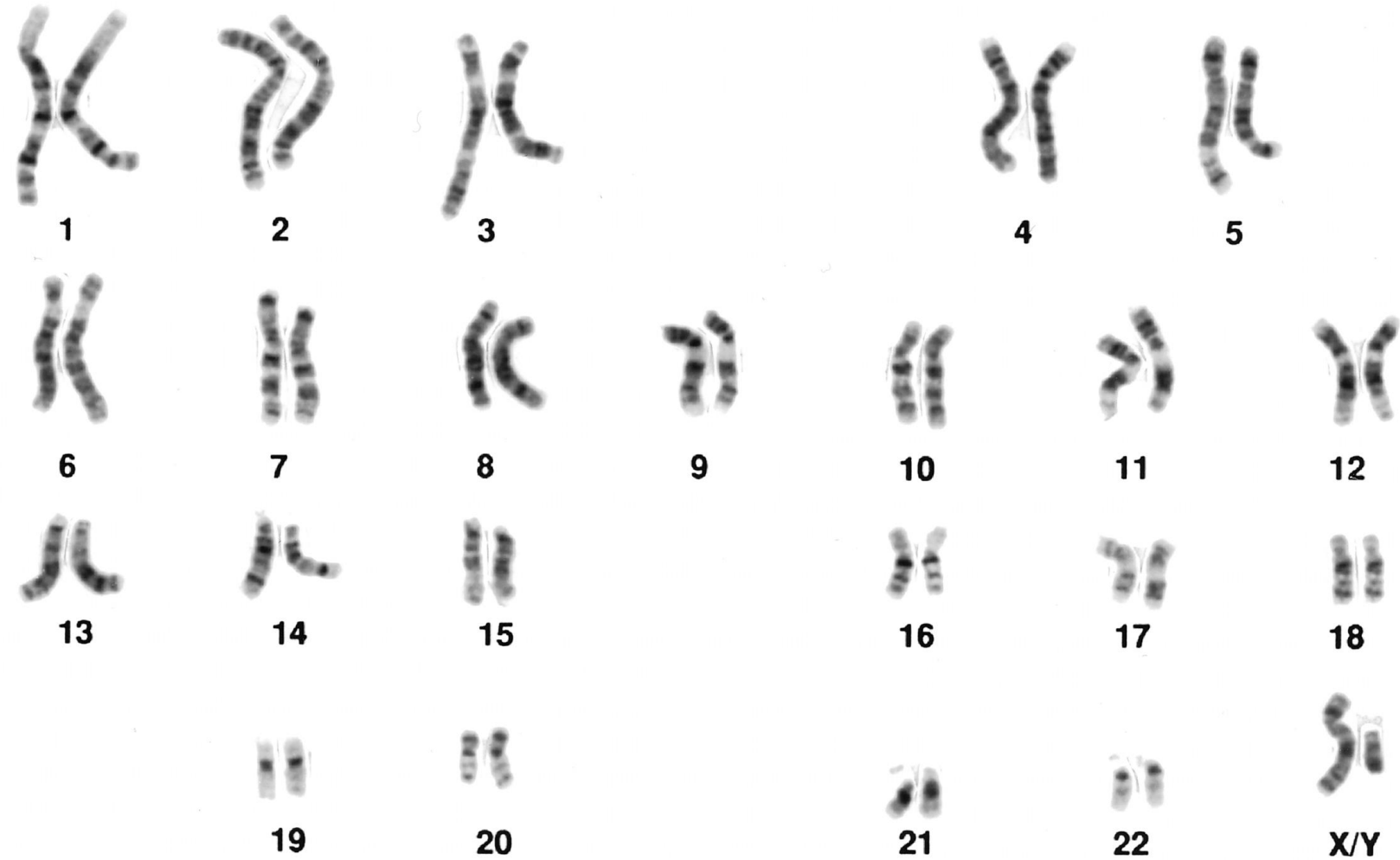
- Normally, blasts represent less than 5 % of BM cells.
- In acute leukemia, this percentage increased to more than normal.
- According to WHO classification, a finding of at least 20 % blasts in the BM is required for diagnosis of the majority of acute leukemias.
- WHO classification depends on the cytogenetics and molecular findings.
- According to FAB classification, a finding of at least 30 % blasts in the BM is required for diagnosis of acute leukemias.
- FAB classification depends on the morphology of blasts, Cytochemistry, and Immunophenotyping (Flow cytometry).

# Diagnostic methods used to study hematopoietic malignancy

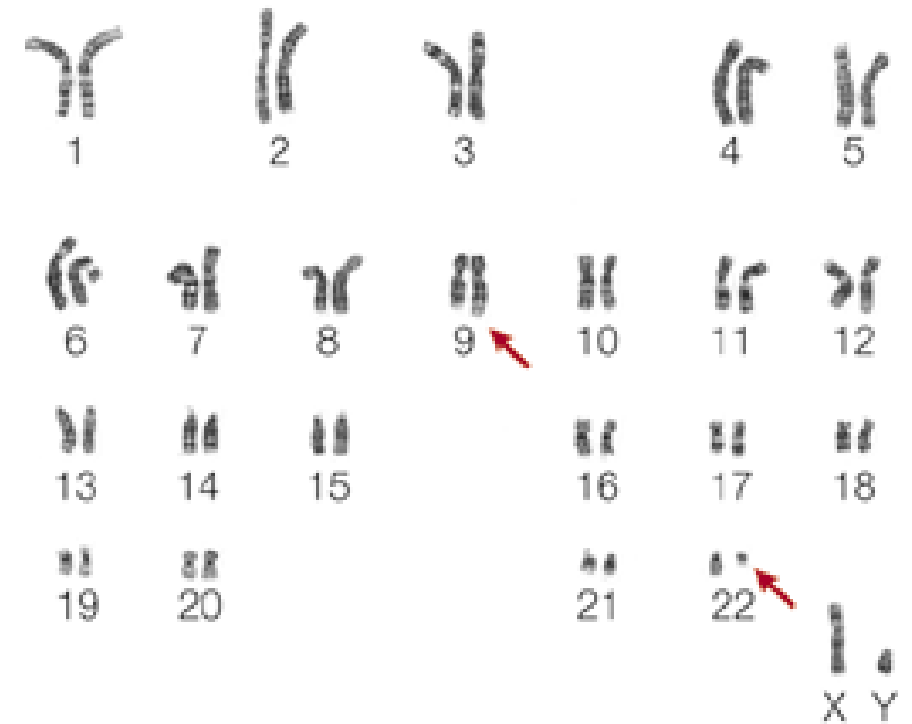
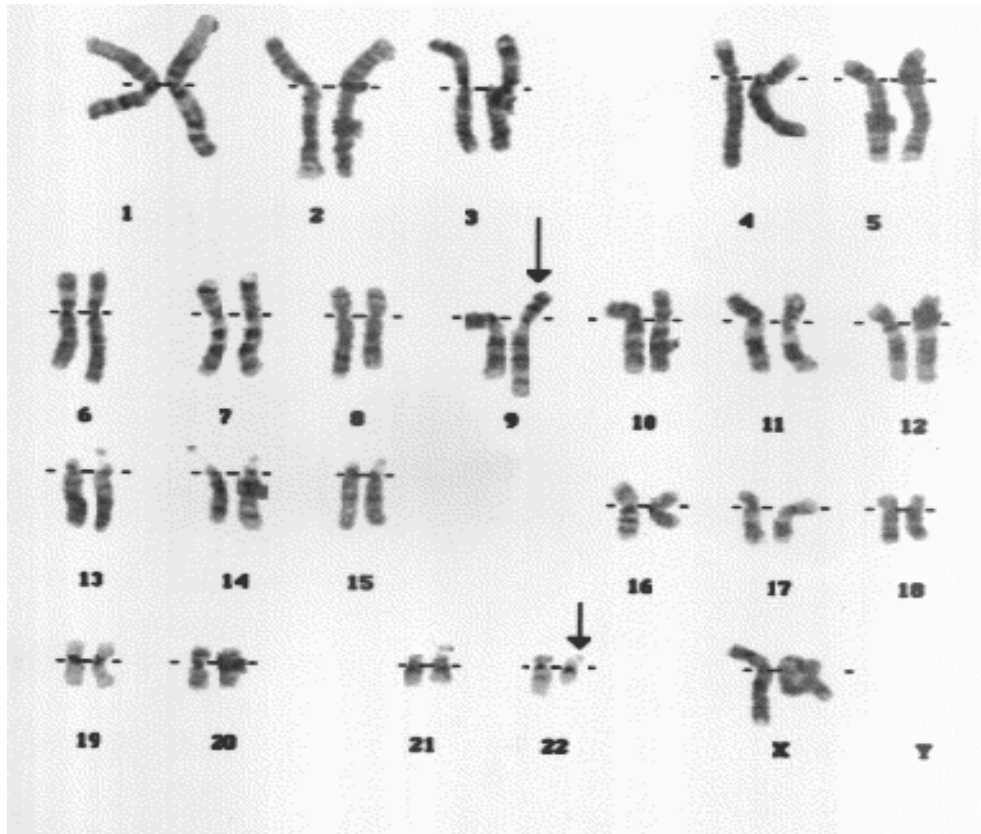
## 1. Cytogenetics (Karyotype analysis):

- Detect genetic abnormalities (numeric and structural).
- Help determine the diagnosis, selection of appropriate therapy, and to expect the disease prognosis.
- Structural abnormalities include Translocations (caused by rearrangement of parts between non-homologous chromosomes), Deletions (in which a part of a chromosome or a sequence of DNA is lost), Inversions (where a segment of a chromosome is reversed end-to-end).

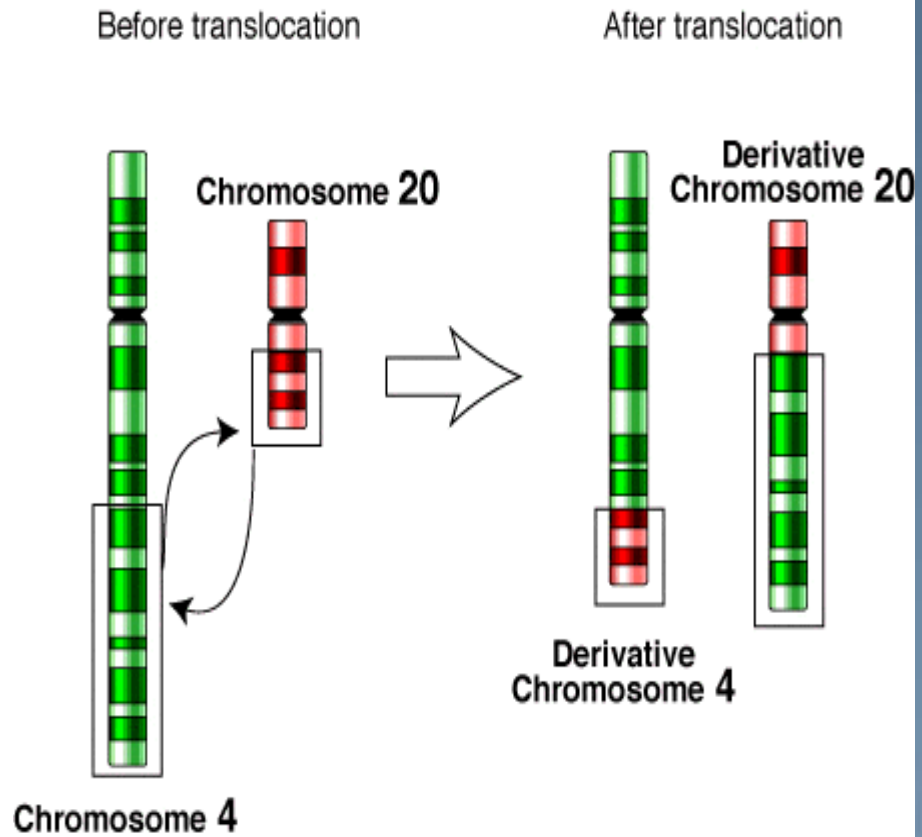




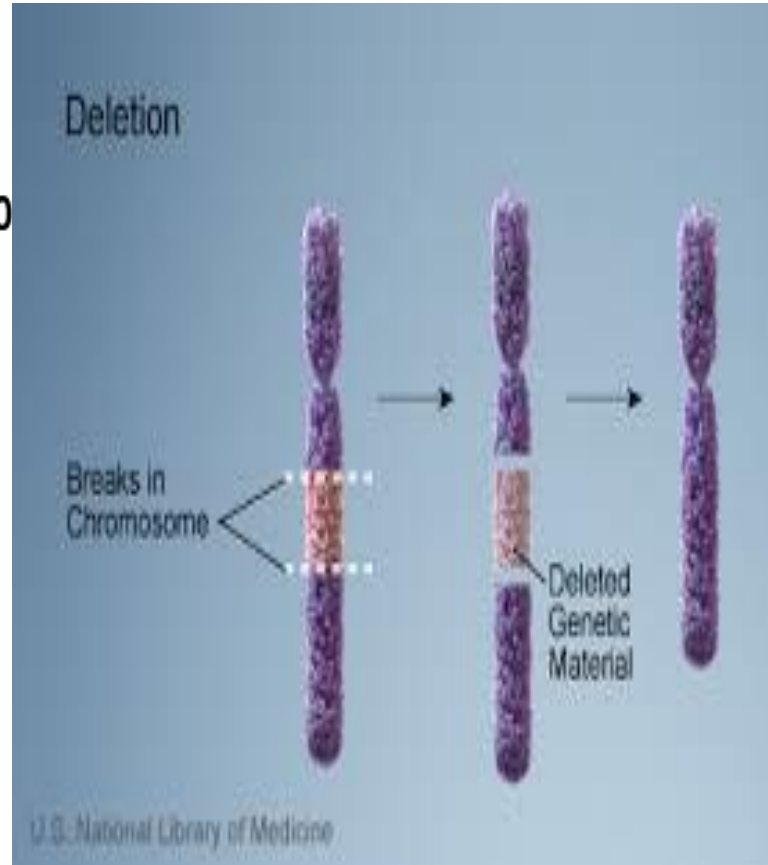
**Normal Karyotype**



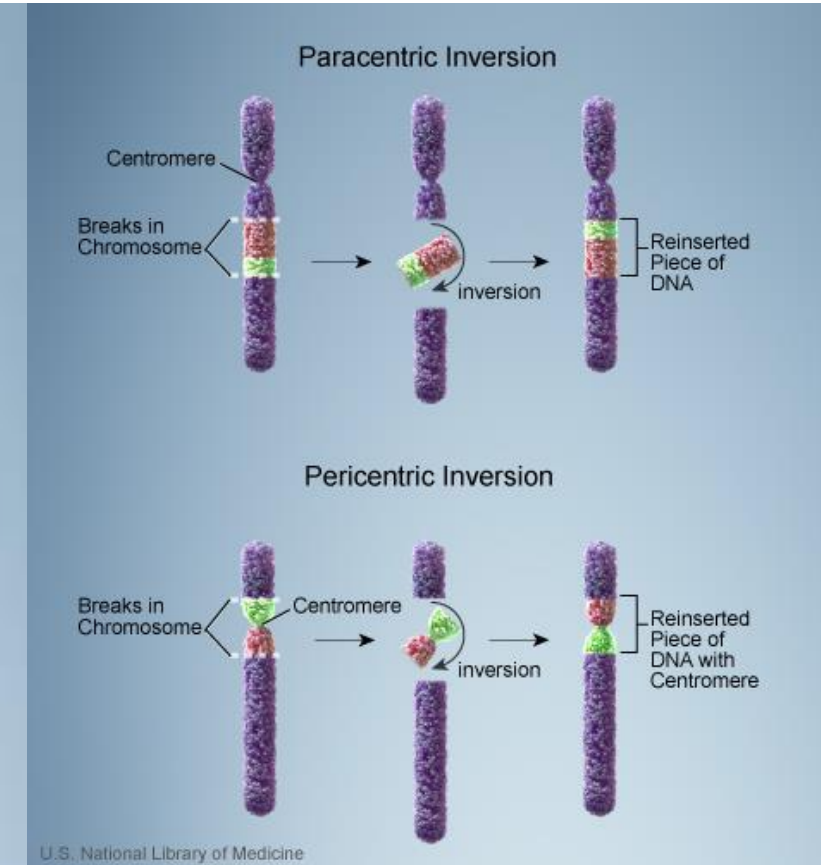
The figures shows 2 patients with t(9; 22) in CML



**Chromosomal Translocation**

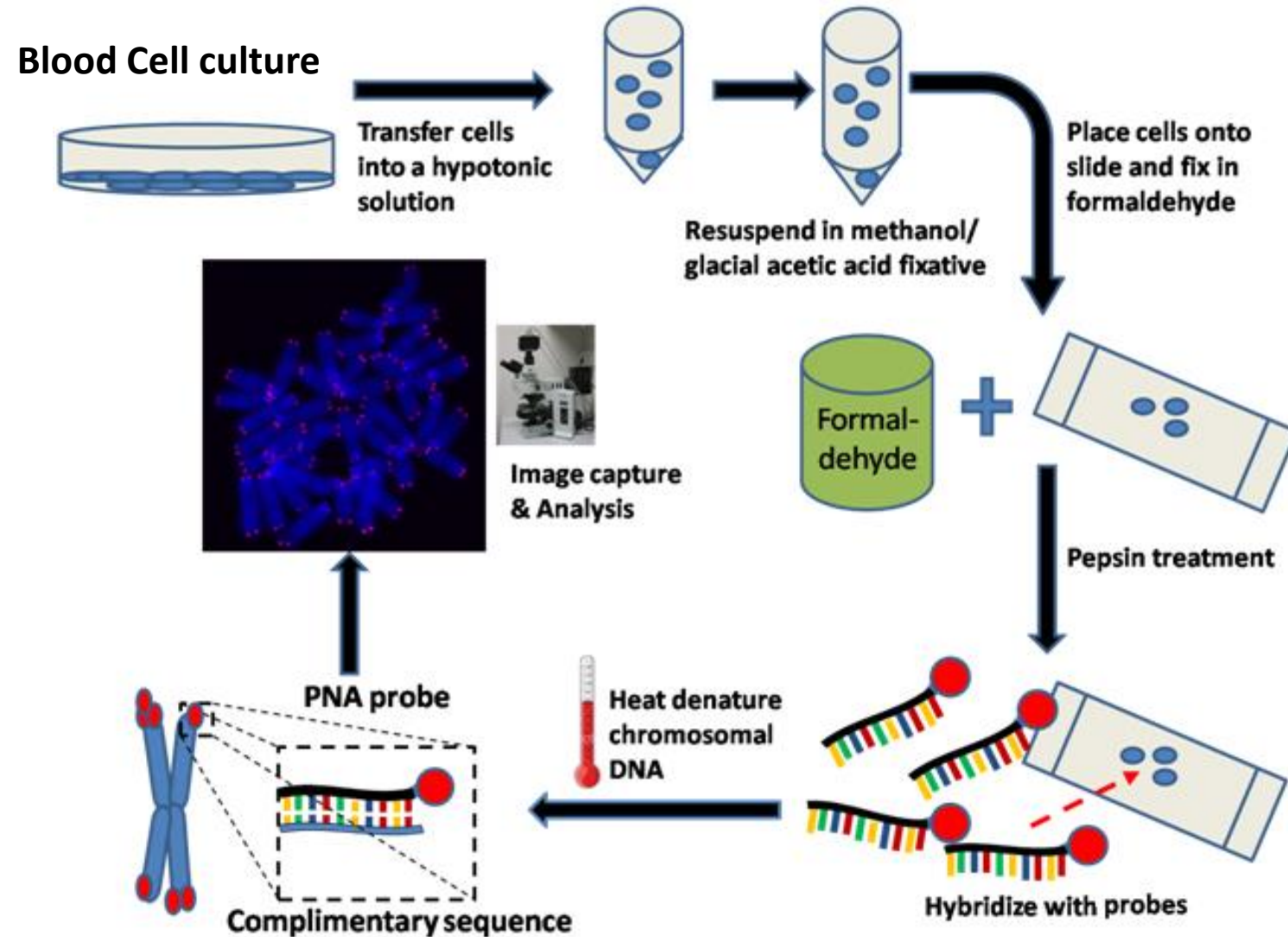


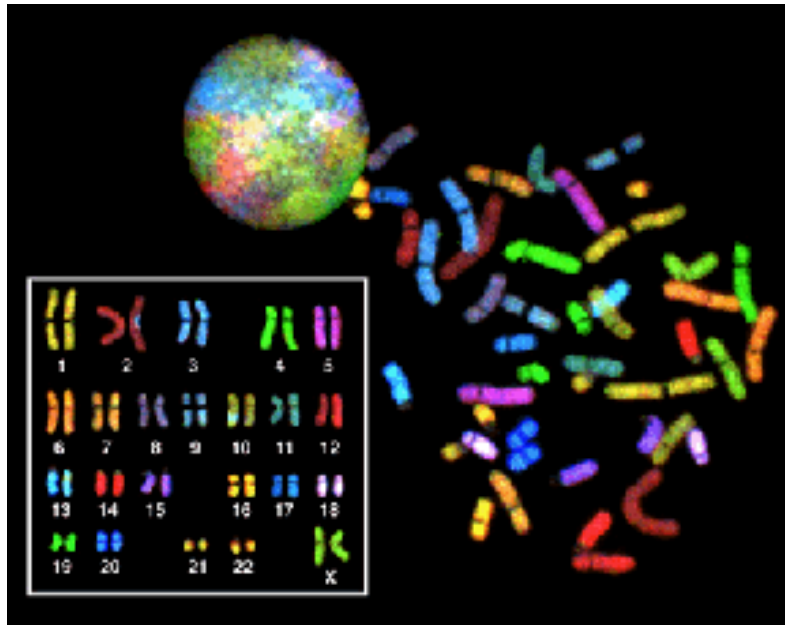
**Deletion**



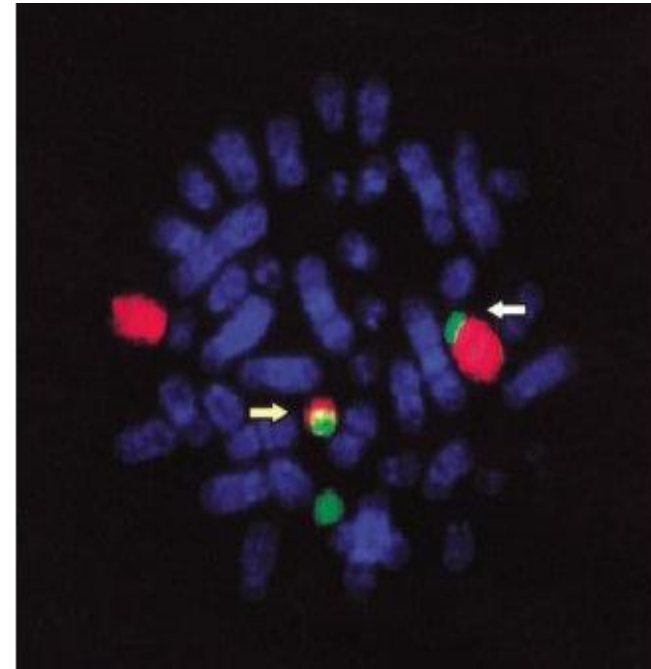
**Inversion**

## 2. FISH (fluorescent in situ hybridization):

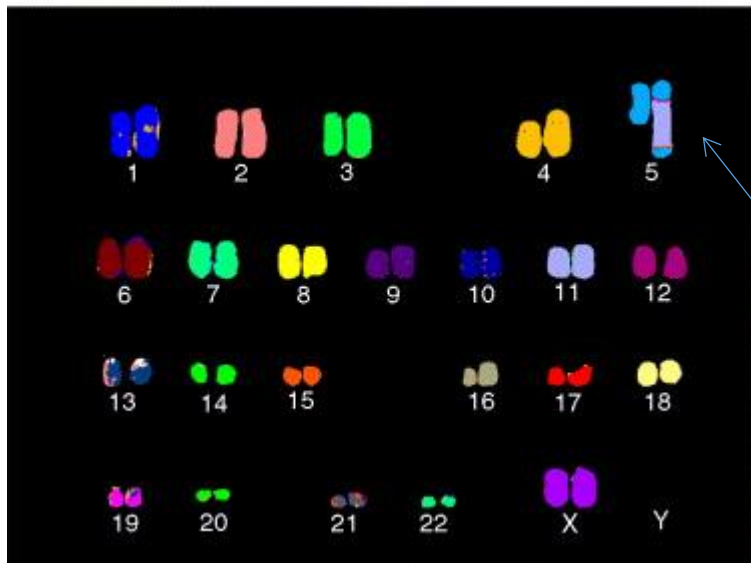




Normal FISH



**t(12; 21) in B- ALL (TEL-AML1)**  
using specific probe targeted the  
translocation site



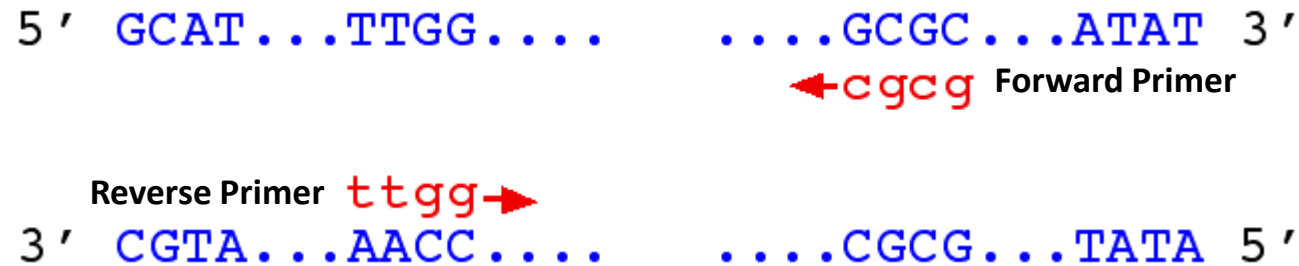
Deletion of chromosome 5 in AML

### 3. PCR (Polymerase chain reaction):

The following components are needed to perform PCR in the laboratory:

- 1) A heat-stable DNA Polymerase (like Taq Polymerase).
- 2) DNA.
- 3) PCR Buffers.
- 4)  $\text{MgCl}_2$  (magnesium chloride).
- 5) All four nucleotide triphosphates, dNTPs (A, T, C, and G).
- 6) Two short, single-stranded DNA molecules that serve as primers.
- 7) Thermal cycler (a device that can change temperatures dramatically in a very short period of time).

- Primers are two short, single-stranded DNA molecules that serve as primers (a short DNA sequence specific to bind to the target gene of interest).
- Forward and reverse primer should be obtained for the reaction.



The sequence length that the forward and reverse primer covers has to be known to check the result by gel electrophoresis

# The three main steps of PCR

- In a PCR reaction, the following steps are repeated 20-40 times.

(note: 25 cycles usually take about 2 hours and amplify the DNA fragment of interest 100,000 fold).

## Step 1: Denature DNA (**Denaturation step**)

At 95°C, the DNA is denatured (i.e. the two strands are separated)

## Step 2: Primers Anneal (**Annealing step**)

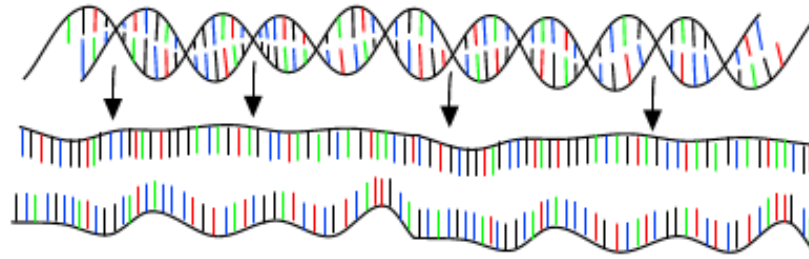
At 40°C- 65°C, the primers anneal (or bind to) their complementary sequences on the single strands of DNA

## Step 3: DNA polymerase Extends the DNA chain (**Extension step**)

At 72°C, DNA Polymerase extends the DNA chain by adding nucleotides to the 3' ends of the primers.

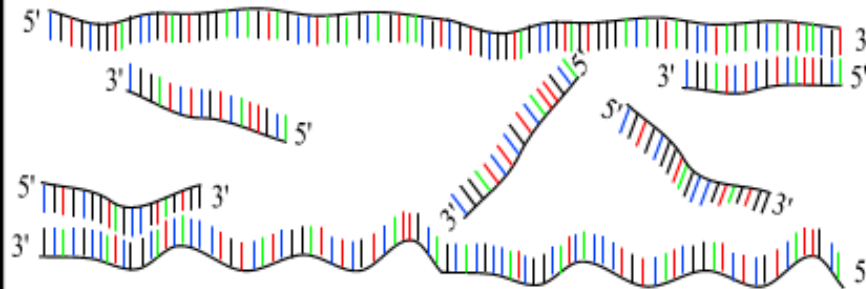
# PCR : Polymerase Chain Reaction

30 - 40 cycles of 3 steps :



Step 1 : denaturation

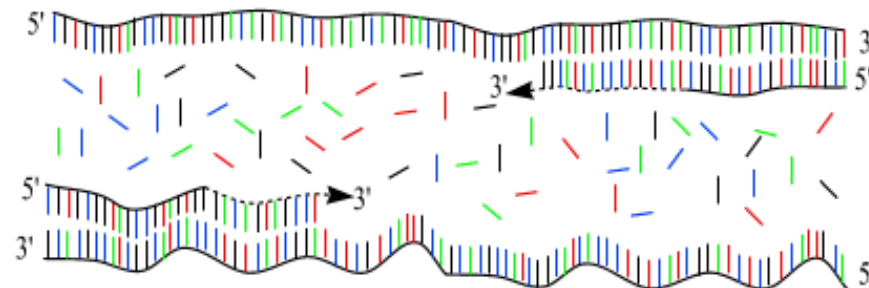
1 minut 94 °C



Step 2 : annealing

45 seconds 54 °C

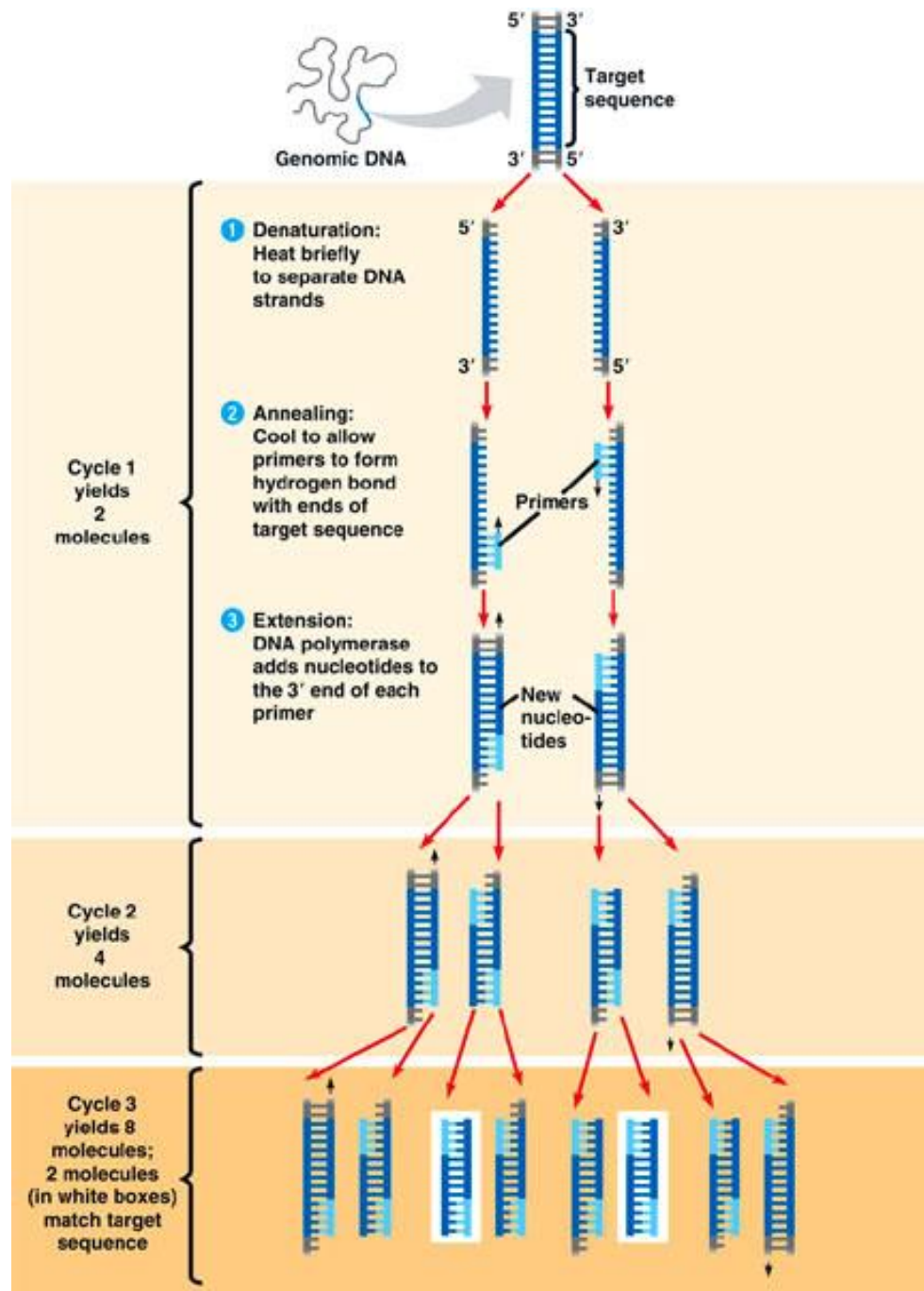
forward and reverse  
primers !!!



Step 3 : extension

2 minutes 72 °C

only dNTP's



Cycling: the generated copies are doubled each cycle.

The DNA of interest is amplified by a power of 2 for each PCR cycle.

For example, if you subject your DNA of interest to 5 cycles of PCR, you will end up with  $2^5$  (or 64) copies of DNA.

Usually, PCR is performed for 30 cycles which means  $2^{30}$   
Means 1,073,741,824.

## Result check at the end of cycling:

- Agarose gel electrophoresis is used.



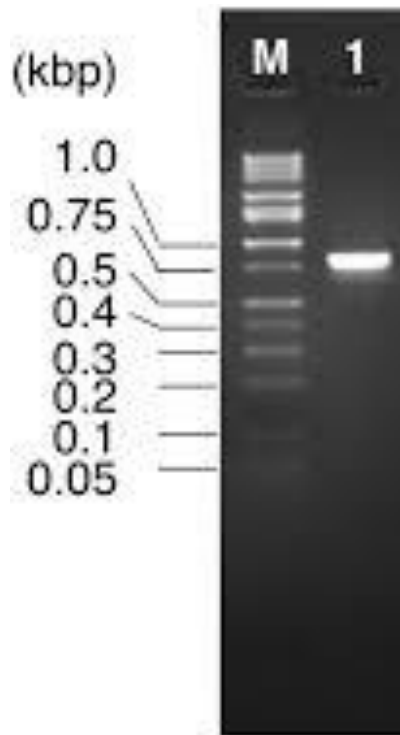
Agarose is mixed with electrophoresis buffer and then heated until fully melting and mixed with **Ethidium bromide**

Agarose gel was then poured into a casting tray containing a sample comb and allowed to solidify at room temperature

pipette into the sample wells a small fraction of the PCR product (the amplified target sequence)

Place the lid on the apparatus, and apply electrical current **DNA will migrate towards the positive electrode.**

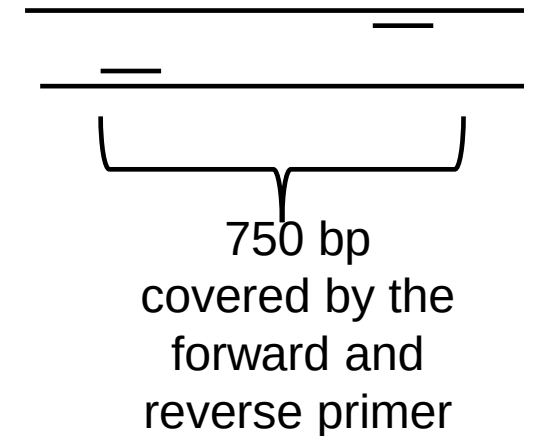
- DNA fragments will migrate in the agarose parallel to the reference-sized ladder (DNA marker).
- Ex. If the amplified fragment by PCR is 300pb it will migrate with the 300pb band in the ladder (bp: base pair).
- **Ethidium bromide**, a fluorescent dye used for staining nucleic acids. DAN will appear as a band on gel when exposed to UV light.



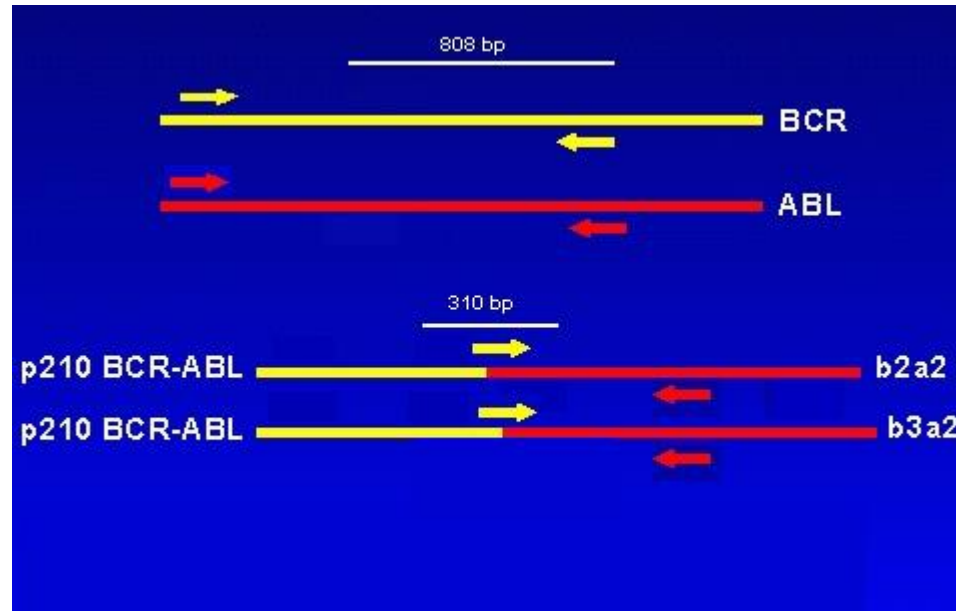
M: DNA marker  
1: is the PCR product

The PCR product in the picture showed that sample 1 migrated with the size 750 bp band (0.75Kbp) this means that the PCR amplified sequence has a 500 bp length.

If the forward and reverse primer flanks the 500bp target sequence it means that the PCR reaction is done correctly.



If we design a primer that targets the translocation sequence we will be able to amplify a piece of the translocated junction for diagnosis.

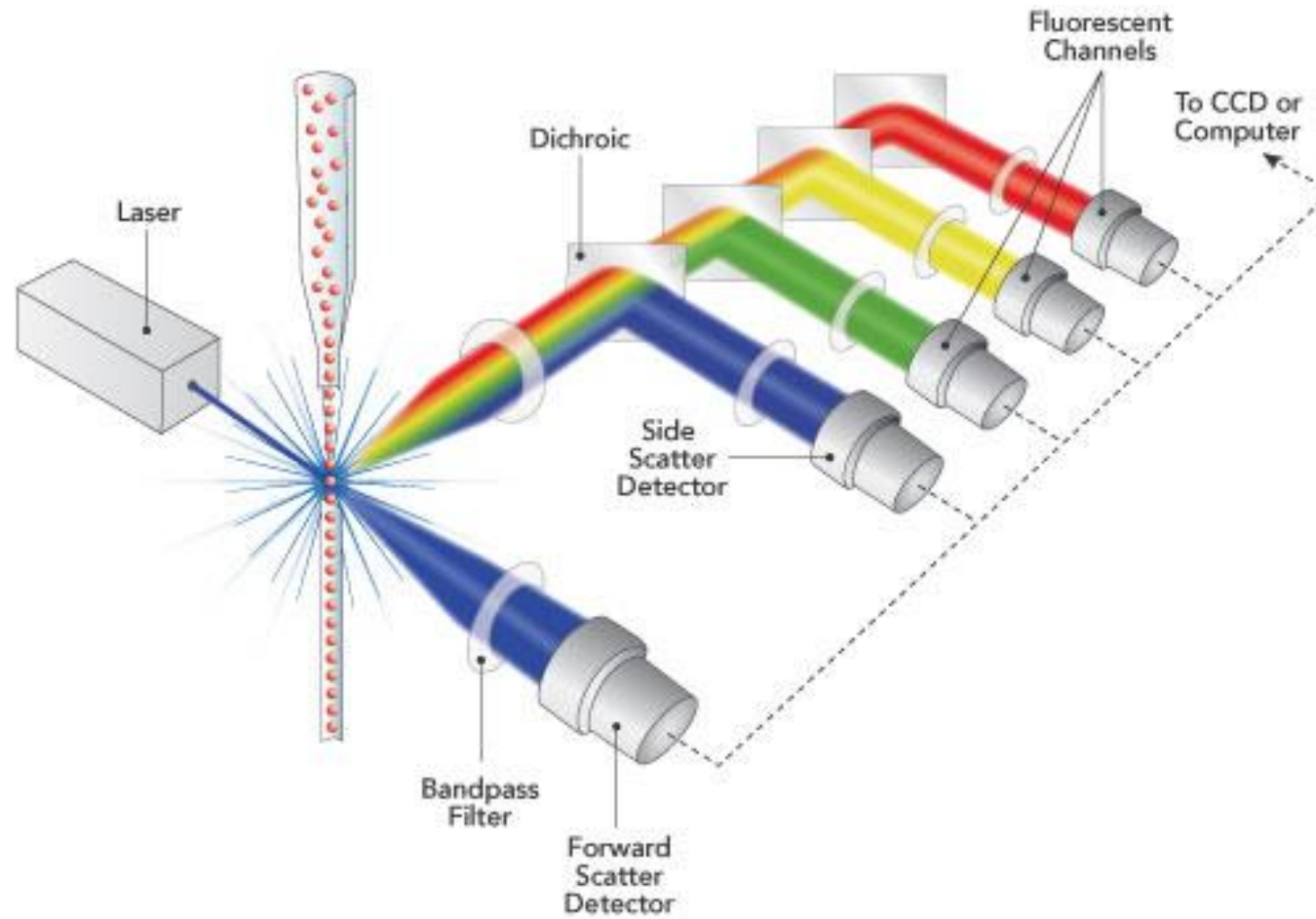


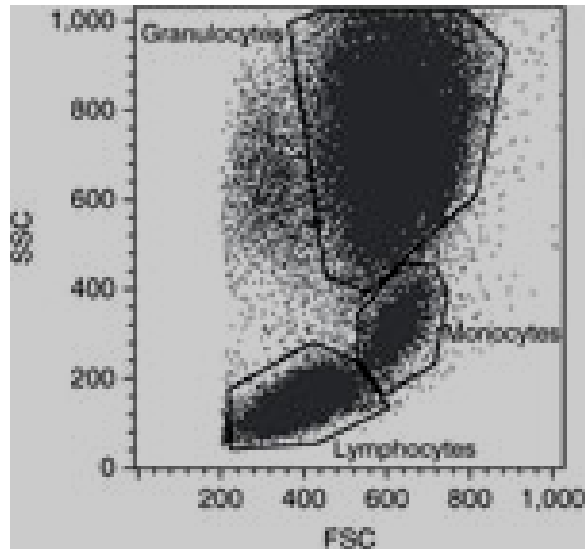
CML diagnosis by PCR



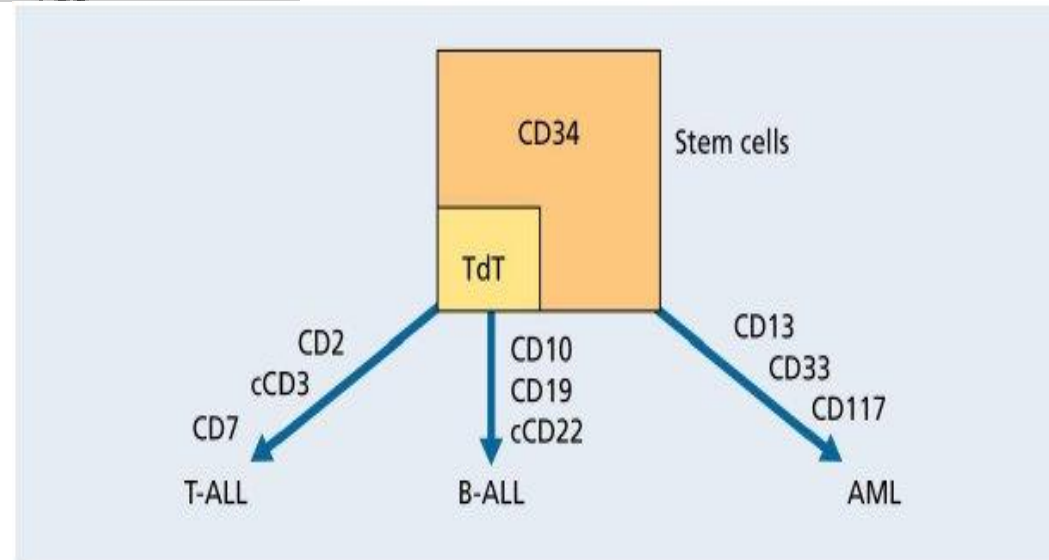
100bp  
ladder

## 4. Flow cytometry:





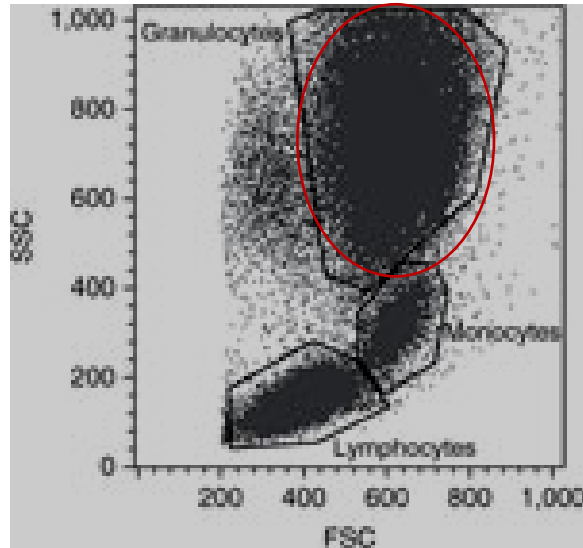
In this technique antibodies labeled with different fluorochrome recognize the pattern and intensity of expression of different antigens on the surface of normal and leukemic cells. This will allow the detection of different types of leukemia.



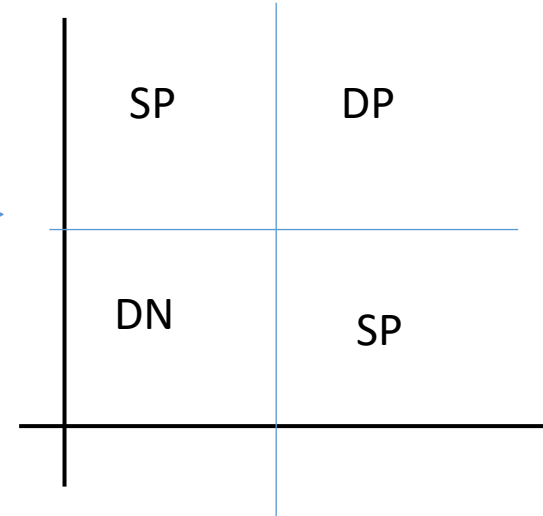
From: *Essential Haematology*, 6th Edn. © A. V. Hoffbrand & P. A. H. Moss.  
Published 2011 by Blackwell Publishing Ltd.

# FACS analysis (fluorescent activated cell sorting analysis on flow cytometry staining)

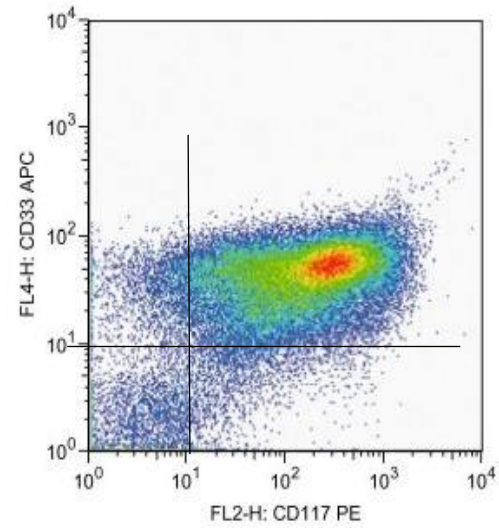
Determine the area to be analyzed



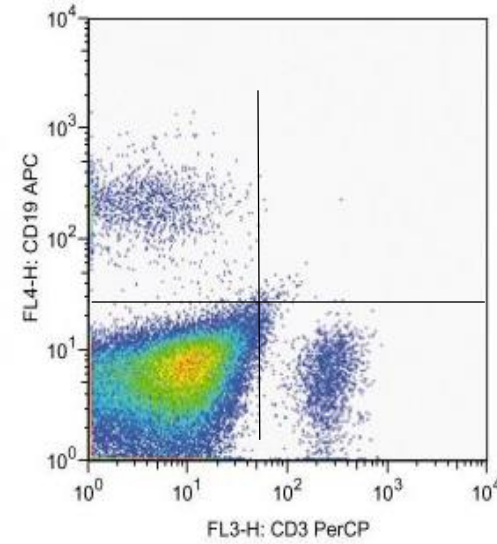
Choose the markers you  
used in the reaction →



# AML diagnosis by flow cytometry



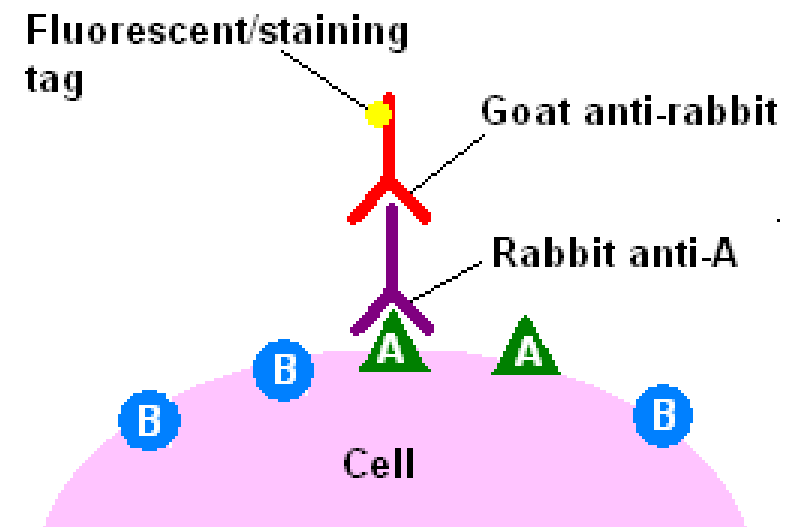
CD33 and CD117 positive



CD19 and CD3 Negative

## 5. Immunohistology:

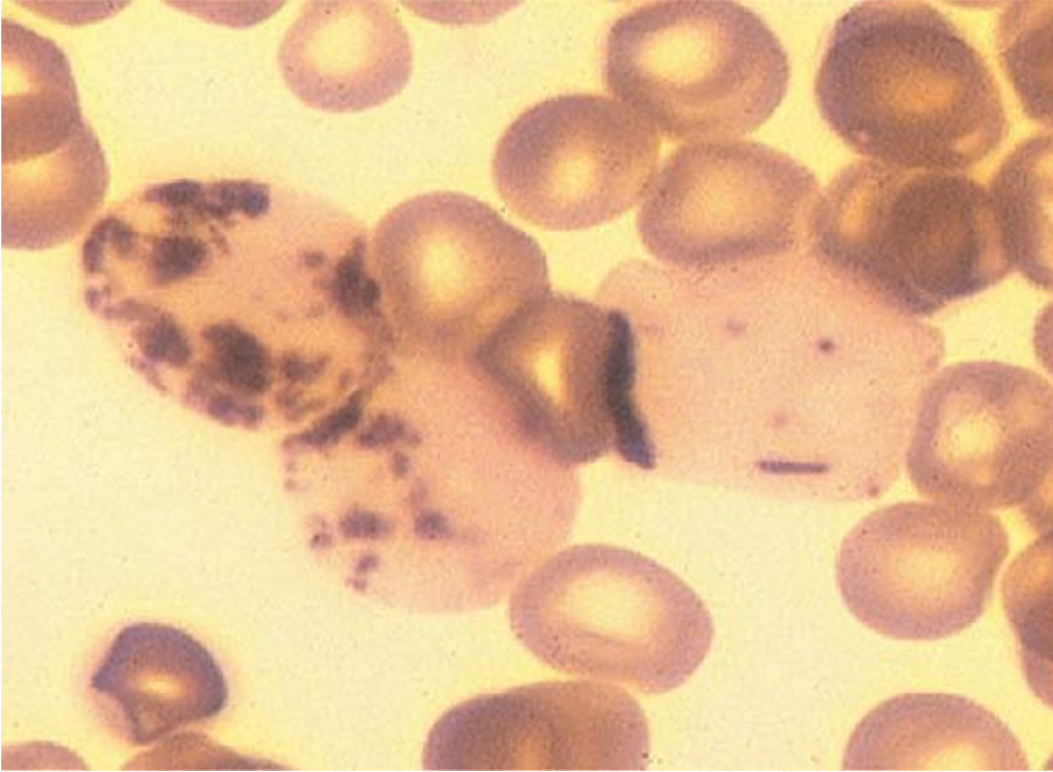
- An antibody can be used to stain tissue sections with a fluorescent marker. Certain receptors or proteins are expressed in cancer cells but not in normal cells.
- Using Abs to detect markers (surface, cytoplasmic Ig, or TdT) associated with cell lineage and maturation stage.
- This can be used in the stain tissue section to help with diagnosis.
- Help to classify AML and ALL and their subtypes.



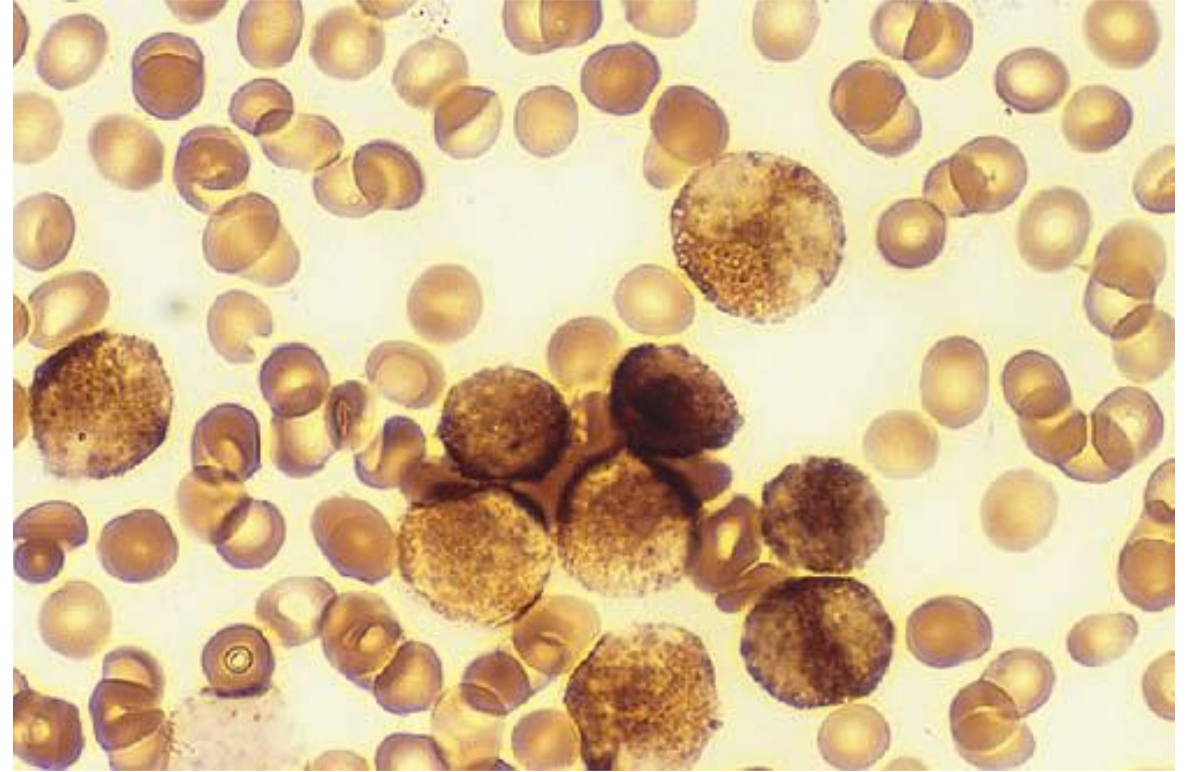
## 6. Cytochemical stain

### A. Myeloperoxidase (MPO)

- MPO is an enzyme found in the primary granules of granulocytic cells and to a certain extent, monocytes.
- Lymphocytes do not contain MPO.
- This stain is useful for differentiating the blasts of acute myeloid leukemia (AML) from those of acute lymphoblastic leukemia (ALL).
- It is positive in AML, and negative in ALL.



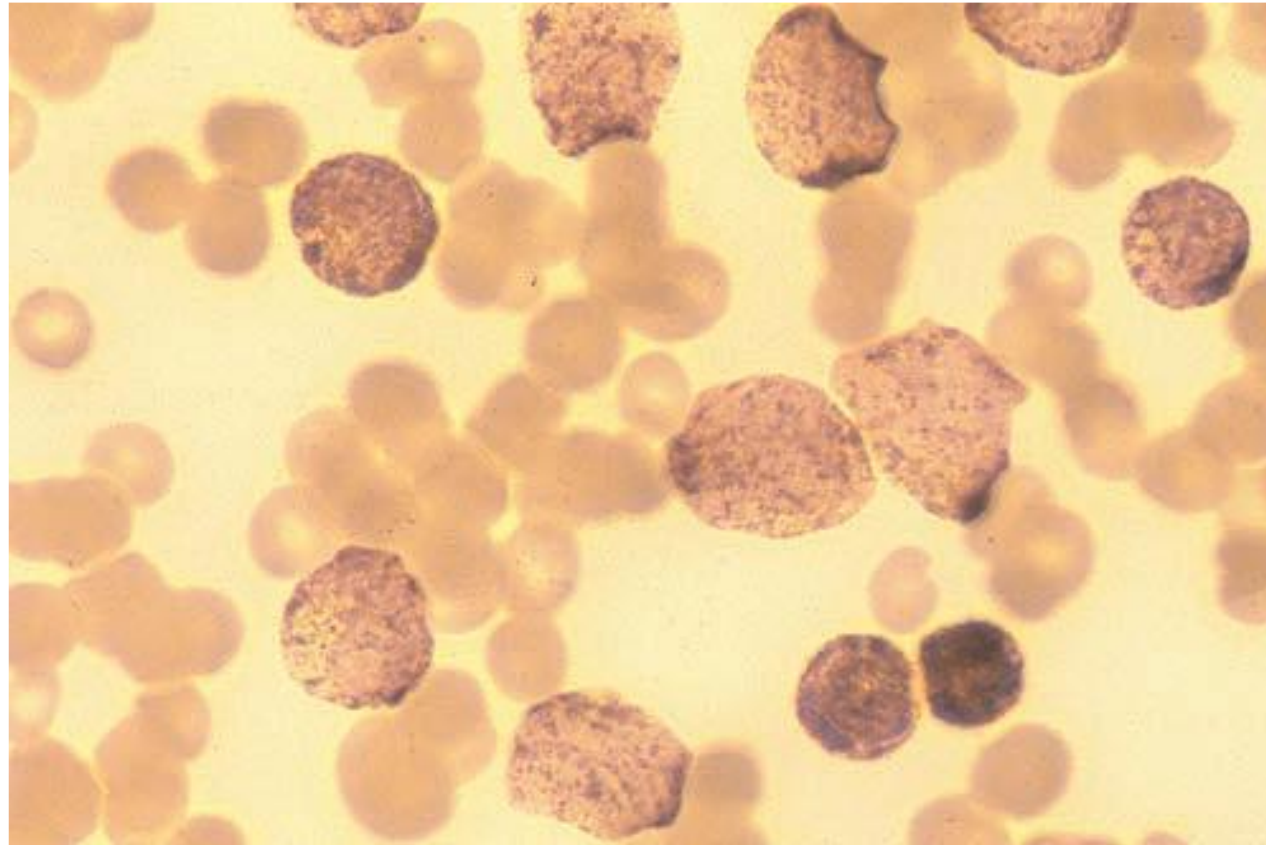
Positive reaction to myeloperoxidase stain in early myeloid cells.



Strong positive reaction to myeloperoxidase stain in leukemic promyelocytes from a patient with acute promyelocytic leukemia.

## **B. Sudan Black B Stain (SBB)**

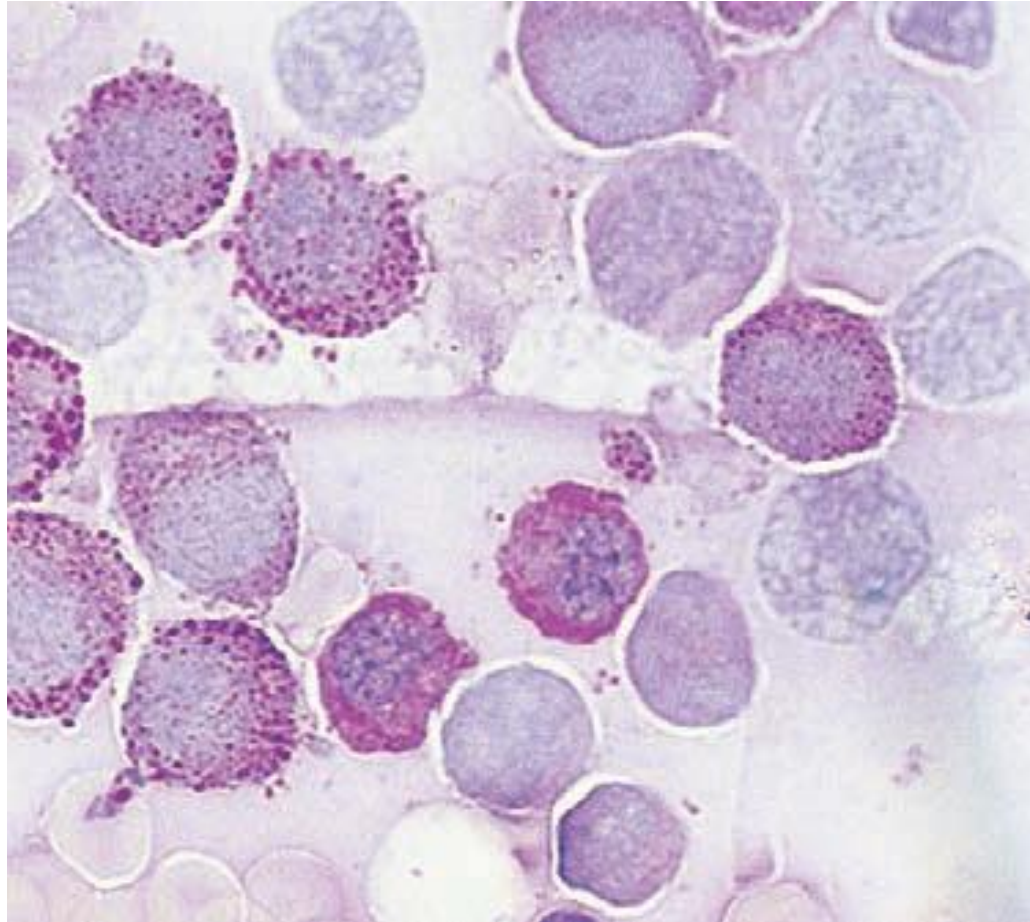
- SBB is possibly a little more sensitive for the early myeloid.
- This stain is useful for differentiating AML from ALL.
- SBB stains lipids such as sterols, neutral fats, and phospholipids, because of its solubility in lipid particles.
- These lipids are found in primary and secondary granules of neutrophils and the lysosomal granules of monocytes.
- It is positive in AML, and negative in ALL.



Sudan black B reaction. The positivity increases with the maturity of the granulocytic cell

## **C. Periodic Acid-Schiff Stain (PAS Stain)**

- A cytochemical stain for complex carbohydrates including glycogen.
- This method is based on the oxidation of glycogen, mucoproteins, and other high-molecular-weight carbohydrates.
- It is helpful in the diagnosis of some ALLs and the erythroid type of AML (M6), as well as in identifying abnormal erythroid precursors in myelodysplastic syndromes (MDS).



Positive periodic acid–Schiff stain

## D. Esterases

Are used to differentiate myeloblasts and neutrophilic granulocytes from cells of monocytic origin.

➤  **$\alpha$  naphthyl acetate esterase (ANAE)**

➤  **$\alpha$  naphthyl butyrate esterase (ANBE)**



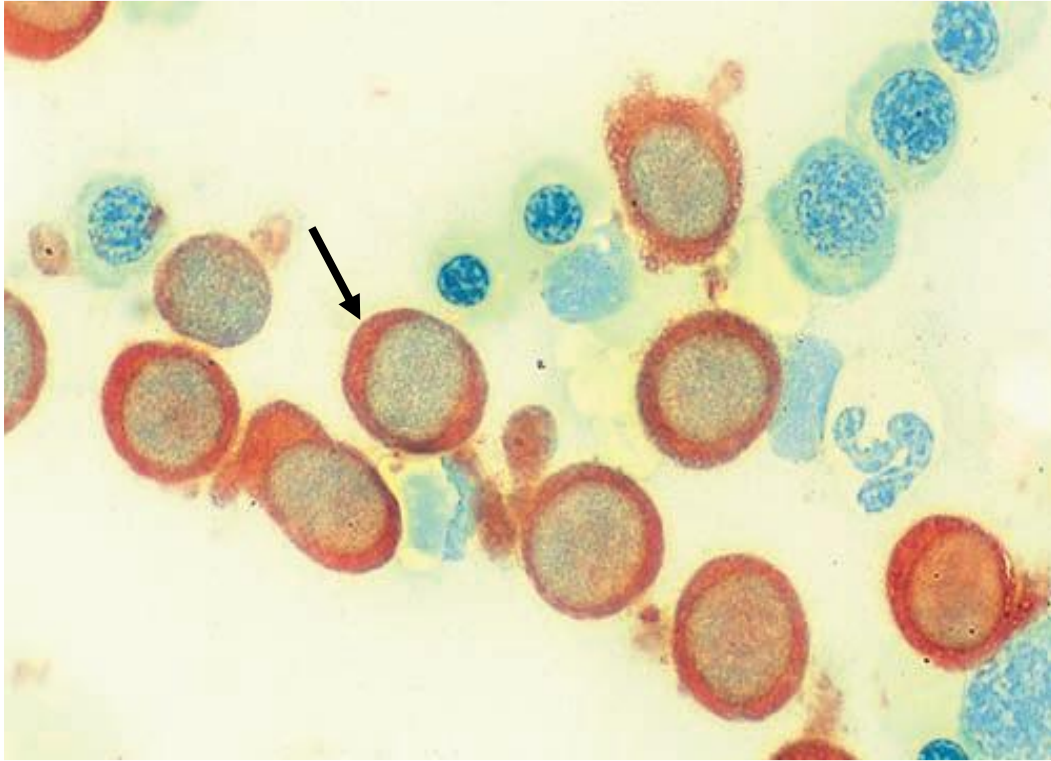
**Non-Specific Esterases**

➤  **$\alpha$  naphthyl chloroacetate esterase (ANCAE) (Specific Esterase)**

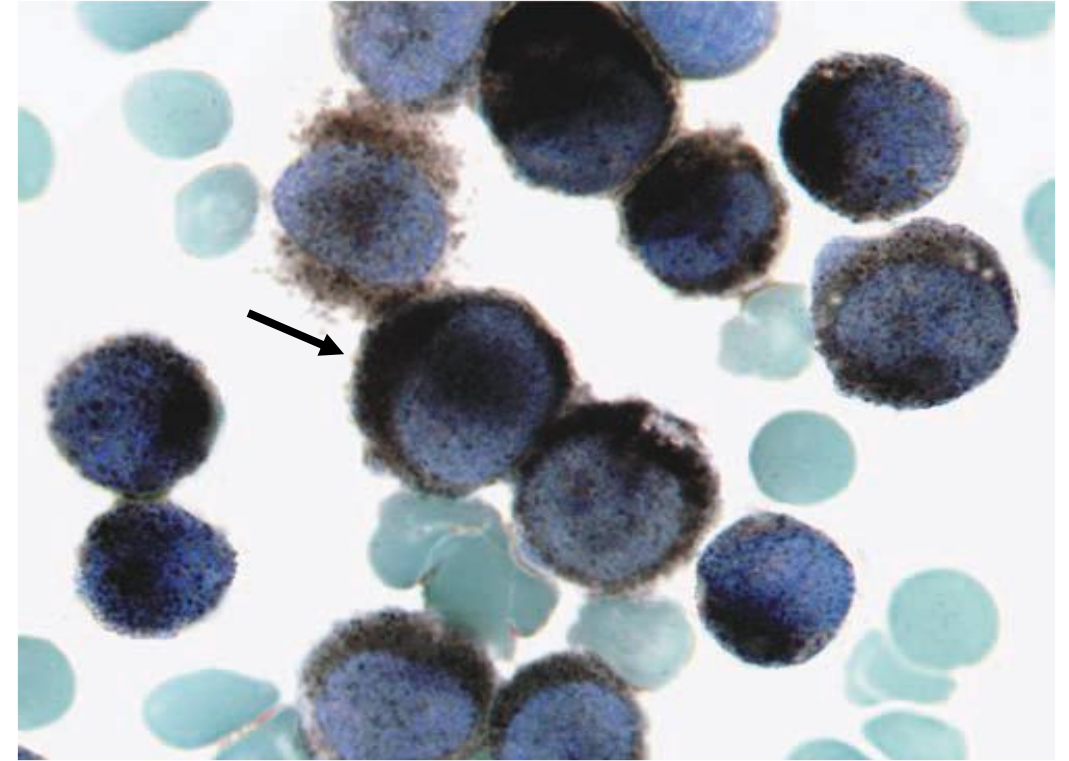
➤ **Non-specific Esterases (NSE)** are a group of enzymes, strongly expressed in cells of the monocytic lineage.

➤ **Specific esterase** is strongly expressed in granulocytic.

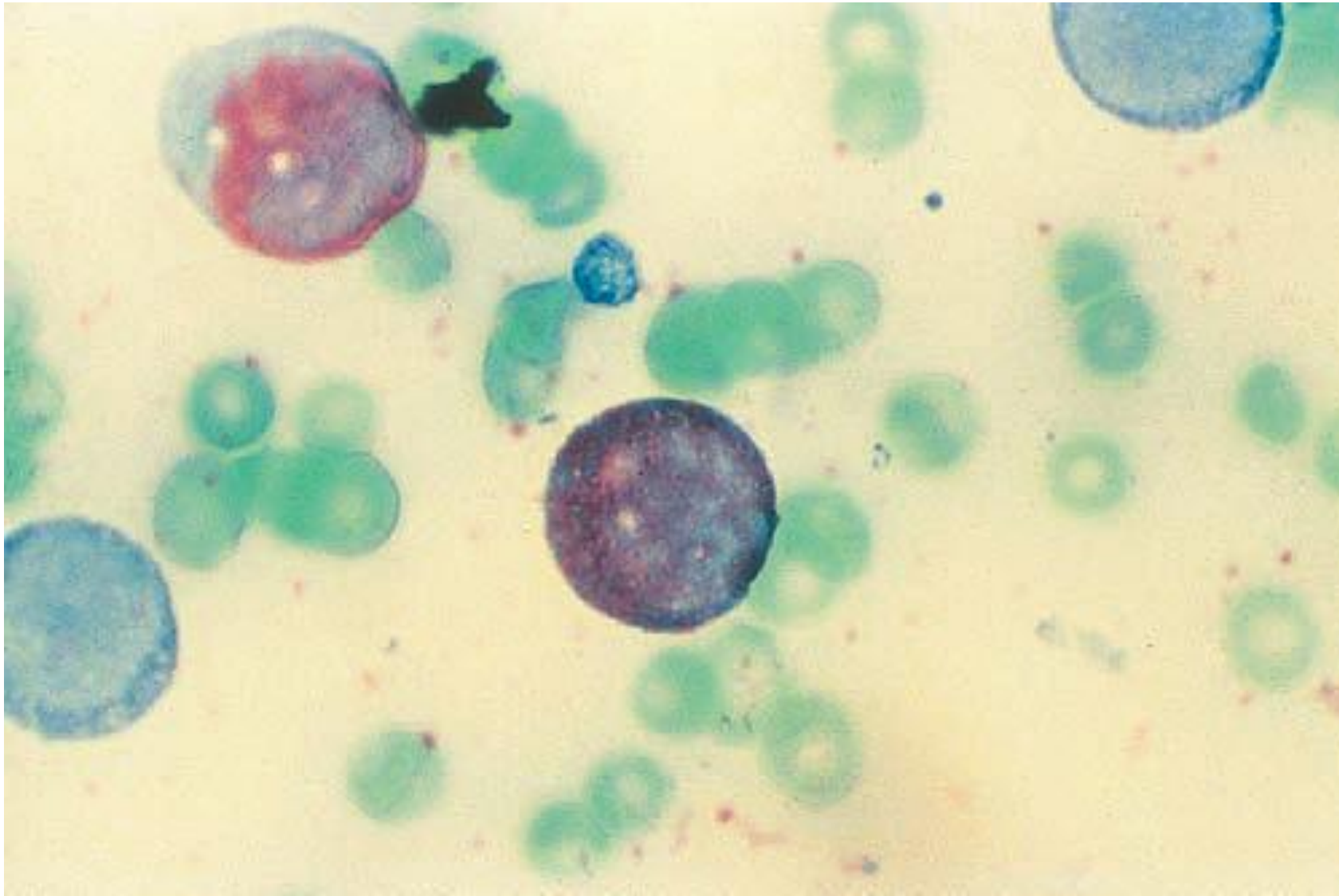
➤ Esterase stains can be used to distinguish acute leukemias that are granulocytic from leukemias that are primarily of monocytic origin.



$\alpha$ -Naphthyl butyrate esterase positivity in cells of monocytic origin from a patient with acute monoblastic/ monocytic leukemia.



Positive reaction to  $\alpha$ -naphthyl acetate esterase stain in monocytes



Positive reaction to AS-D chloroacetate esterase stain in two granulocytic cells

|                                    | ALL                                | AML                     |
|------------------------------------|------------------------------------|-------------------------|
| <b>Myeloperoxidase (MPO)</b>       | -                                  | + (including Auer rods) |
| <b>Sudan black (SBB)</b>           | -                                  | + (including Auer rods) |
| <b>Non-Specific Esterase (NSE)</b> | -                                  | + in M4,M5              |
| <b>Periodic Acid Schiff (PAS)</b>  | + (coarse block positively in ALL) | + (fine blocks in M6)   |